

LangGraph State Machine Architecture & Cyclic Workflow Guide

1. Overview of LangGraph

LangGraph enables building robust, stateful, multi-actor AI workflows as directed cyclic graphs, overcoming the linear limitations of standard chains.

2. Core Concepts

Concept	Definition in LangGraph
State	Central typed dictionary (TypedDict) passed and mutated across nodes
Nodes	Python functions or LLM calls that receive and update state
Edges	Directional transitions between execution nodes
Conditional Edges	Dynamic routers that branch execution paths based on LLM outputs
End State (END)	Explicit termination node preventing runaway loops